
100X
iQOO

ShowHow

Capture once. Learn anywhere. Work without the internet.

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Problem

- Most practical work knowledge exists only in people's heads. It includes the **right order of steps, important checks, and small corrections** that are hard to write down.
- Creating manuals takes too much time, while watching videos is difficult when your **hands are busy**. Both require people to stop their work and focus on a screen.
- So, most people rely on **live demonstrations,** which have to be repeated for every new person. Nothing gets properly recorded.
- **886M Indians are online, but 98% use Indian languages.** Yet guides remain English-first, while voice assistants struggle in noisy, hands-busy environments.
- As a result, this valuable knowledge remains **undocumented, hard to find, and dependent on one experienced person.**

The Core Gap

TODAY

Expert demonstrates
Knowledge lives in their head.

Learner follows
Needs to stop and pay attention.

Expert repeats
Same explanation, again and again.

WHAT IS MISSING

Capture the demonstration
No easy way to capture the process automatically

Guide the process hands-free
No way to follow instructions without stopping work.

Make the knowledge reusable
No system to store and reuse what was taught.

RESULT

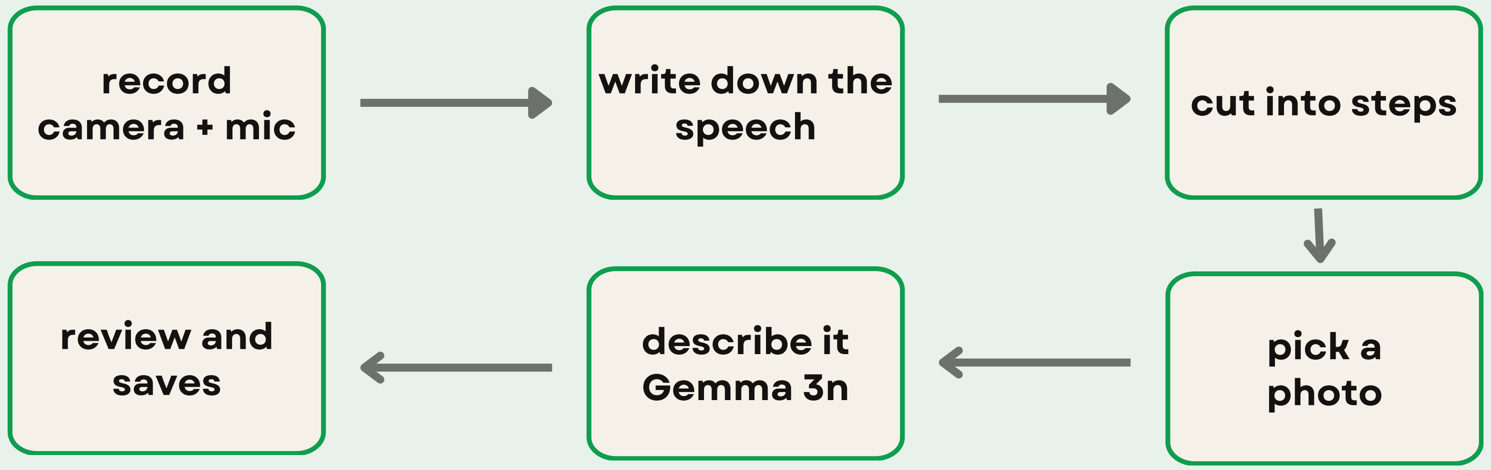
Knowledge stays with the expert only.

Solution

ShowHow is one app with two modes. The person who knows the job does it once and talks through it. Everyone after that is taught by the phone – hands-free, in that person's own voice, in their own language, with no internet and everything locally.

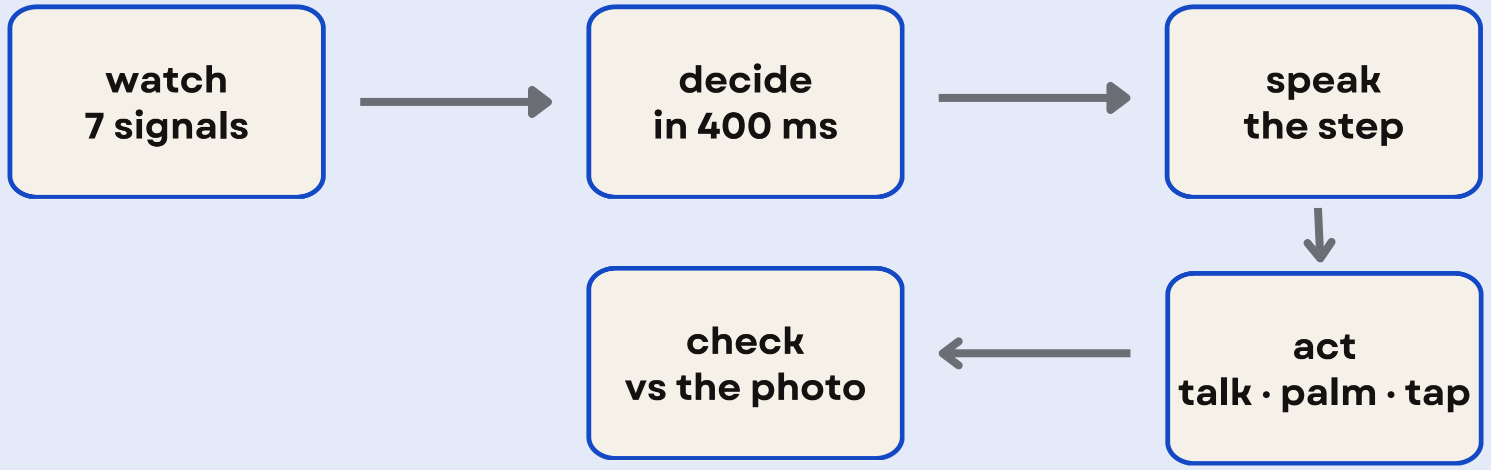
SHOW once, by the person who knows

- An experienced person **does the job once and explains it aloud** while the phone **watches and listens without being touched.**
- AI understands the explanation and **breaks the work into steps**, saving **one photo and the original voice** for each step.
- In about **90 seconds**, the demonstration becomes a complete guide.



GUIDE after that, for everyone else

- The next person starts working with the phone nearby.
- The phone watches **7 signals** to understand what is happening and automatically decides whether to **speak, watch, or wait for a tap.**
- It gives instructions **hands-free in the original person's voice** and checks whether the current setup matches the correct step, warning the user if something is wrong.



Phone - First.

The iQOO 15 is the camera, the writer, the teacher and the checker. **Four jobs, one phone**. Every part of ShowHow uses the phone's camera, microphone or AI chip. Take the phone away and there is nothing left to show.

Office Kit	Four real uses, plus the build. **Remote control lets the laptop keyboard fix step wording at desk speed.** File transfer moves **4 MB guide folders in and out**. Screen mirror puts one phone on a projector for a training room. The shared clipboard carries the live run log across. And every rule and word list we tune is a text file dropped over Office Kit – so we use it all weekend, not just in the demo.
Technical depth	We use AI only for language, and simple code for everything else. **The step cutter uses a word list and a timer**. The rules engine uses **40 simple rules stored in a file** that we can open and read. We also show which chip handles each part and clearly mention the four things we cannot measure honestly.
End product quality	**Two judges, one phone, airplane mode, 2.5 minutes.** One records a job; the other, who has never seen it, is taught by the phone. The guide they walk away with did not exist when they sat down – and nothing in that demo is simulated by a presenter.

WHAT WE WILL MEASURE ON THE iQOO 15?

Making a guide
How long it takes to turn a 60–90 second recording into a finished guide, and how long each step takes.

Speed
How quickly the phone changes modes, reacts to a hand movement, and turns speech into an action.

Working offline
Everything runs offline on the phone. We measure its hardware use and confirm no data leaves the device.

The 30 hour Sprint

30 hours total. 19 focused build hours, with 10.5 hours done entirely on the phone. The product does not need a laptop, and our Red Light plan does not need a rebuild. The rules, prompts, word lists and guides are simple text files we can change anytime. We spend Red Light building, not waiting.

WHEN	LIGHT	WHAT WE BUILD	WHY THEN
Sat 11:00–13:00	GREEN	Basic app, sensor setup, and all models running on the phone	Laptop tools are allowed now. Get the heavy files onto the phone first.
Sat 13:00–15:30	RED	Show Mode: camera buffer, live transcript, step maker, and review screen	This is the heart of the product, and it all needs to work on the phone.
Sat 15:30–16:30	GREEN	Breather: speech model and chip issues that need deeper debugging	Our one planned window for anything that really needs a laptop.
Sat 16:30–19:00	RED	Rules, hand signs, and the on-screen reason bar, all tuned from policy.json	Changing a text file means no rebuild. This is exactly what Red Light is for.
Sat 22:00–Sun 01:00	RED	Guide playback: voice clips, four guide layouts, and photo checking	The second half of the loop. We prove it works on the phone.
Sun 01:00–06:30	GREEN	Photo descriptions, Office Kit testing, and a one-hour heat test	Long tests can run while the phone does the work.
Sun 06:30–09:00	RED	Record five real guides. Practice in airplane mode. Break it on purpose.	The demo is phone-only and offline, so we practice it the same way.

How it works?

SHOW — Record once, by the expert (~90 sec of talking, hands stay on work)

1. **Talk** — Camera & mic run. Nothing to press.
→ 2. **Transcribe** — Hindi/Marathi speech is written on the phone.
→ 3. **Cut steps** — Marks: phir, ab, next — adds pauses.
→ 4. **Photo each step** — Best frame + voice clip saved.
→ 5. **Review & save** — Join/split/redo a step. ~20 sec of cleanup.

One folder = the guide (~4 MB) • No server • No account • No upload • Copy to another phone via cable

steps.json • audio/ • photos/ • vectors.bin

GUIDE — Anyone runs it, hands full — phone reads the situation and helps

- **How it's held** — In hand / flat / moving
- **How loud the room** — Mic level (checked often)
- **How clearly I can hear** — Based on last 3 answers
- **How far the user is** — Front camera check (every 2 sec)

→

Simple rules (no AI)

- ~40 rules for these signals
- Waits 400 ms before switching
- Decides in < 10 ms
- All numbers in one file

No AI model in this path

→

TAP	In hand, quiet room Normal cards.
TALK	Phone down/far away Big text + voice reads step, answers back.
HANDS	Loud or unclear speech Camera on → Palm → fist → repeat → 👍
EASY	Chosen in settings Bigger text, slower, confirms first.

Always shows why. A bar at the bottom tells the reason. | Example: HANDS ← room is loud (~20 dBFS)

BUILT WITH

- Flutter • Dart
- Android + Web
- One codebase
- setState • Kotlin channel

MODELS ON DEVICE

- IndicConformer-600M (speech)
- Gemma 3n E2E (captions)
- EmbeddingGemma (photo vectors)
- MediaPipe (hand signs)
- Silero VAD

CHIP DOES WHAT

- Sensor hub: motion, loudness
- NPU / GPU: speech, captions, vectors
- GPU: hand signs, face distance
- CPU: rules, every fallback
- If delegate won't load → drops a tier

WE PROMISE

- No internet permission
- No login, no cloud
- Camera opens only when asked
- Warnings never block the worker

Impact.Usefulness

For the Expert

The **expert records the work once in their own words**, so the same explanation doesn't have to be repeated every time.

For the New User

The steps are available in their language and can be followed **hands-free**, even when the expert is not there.

For the Workplace

Training becomes easier because new people can learn the process without waiting for **someone to explain it again.**

Wider Impact

With **98%** consuming Indian-language content, offline, local-language instructions can help close the gap.

Beyond Workshops

The same system can **support kitchens, garages, labs, farms, and machine setups** wherever the same explanation is repeated.

Feasibility. Team

Why it is feasible?

- **We listen for steps instead of trying to understand the whole video.** People naturally say words like “phir,” “ab,” or “uske baad” while explaining. We use these words and pauses to split the recording into steps. No model to train.
- **The review screen makes it easier**. The user can check and fix the steps before saving. So the cuts only need to be close enough, not perfect.
- **We use the expert’s own voice.** There is **no TTS or voice cloning**. We simply replay the original recorded audio, keeping the natural voice and language.
- **No AI is used for decisions.** Choosing between four modes based on seven signals is handled by a simple state machine, making it fast and predictable.
- For the rest, **we use IndicConformer for speech, Silero for speech detection, and MediaPipe for hand signs**, running on the phone. If one processor is slow, the system can fall back to another.

Who is Building?

KSHITIJ DESAI (leader)

On Device AI, System Design

Owns the pipeline from audio to guide folder, plus the rules engine.

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ANUSHKA UNDE

UI/UX, Frontend, APIs

Builds both screens the judge touches, and the file layer that moves guides to the laptop.

https://www.linkedin.com/in/anushka-unde-a389a3271/

AYAAN MEMON

AI, Leadership, Communication

Owns the language words the app listens for, and the pitch.

https://www.linkedin.com/in/ayaan-memon/

Research

Learning from Narrated Instruction Videos (Alayrac et al.)
Narration and video together reveal a task's steps. → We took the narration half and dropped the model.

Multimodal Context Awareness for Procedural Videos (CHI 2023)
Thirty people, an hour-long cooking task. Following a procedure needs sensing of the room, not better video controls. → Our seven signals.

ARify (CHI 2026)
Experts – technicians, repair professionals, chefs – face real barriers to writing guides. → The problem is current. They need AR hardware; we need a phone.

Internet in India 2024 (IAMAI–Kantar)
886M online, 488M rural, 98% consume in Indian languages. → Instructions still arrive as English text.

Thank you